

Literature status: local diameter 2 property versus diameter 2 property

Run `fa_banach_001`; source arXiv:1206.3539

Source question

Abrahamsen, Lima, and Nygaard’s paper *Almost isometric ideals in Banach spaces* defines, in Section 3, the local diameter 2 property and the diameter 2 property as follows. A Banach space X has the local diameter 2 property if every slice of B_X has diameter 2, and it has the diameter 2 property if every non-empty relatively weakly open subset of B_X has diameter 2. Immediately after Definition 3.1, on page 7 of the PDF, the authors state that they do not know whether these two properties are really different.

Supporting answer

Becerra Guerrero, Lopez-Perez, and Rueda Zoca’s paper *Big slices versus big relatively weakly open subsets in Banach spaces*, arXiv:1304.4397, answers this question negatively. In the introduction, on page 2 of the PDF, they say that the aim of the note is to prove the existence of a Banach space satisfying the slice diameter 2 property and failing the diameter 2 property, answering the open problem negatively.

There is one notation warning. In arXiv:1304.4397, “SD2P” denotes the slice diameter 2 property. This is the same condition as the local diameter 2 property in arXiv:1206.3539. It is not the modern strong diameter 2 property.

The decisive statement is Theorem 2.4, followed by Corollary 2.5, on page 10: every Banach space containing an isomorphic copy of c_0 can be equivalently renormed so that every slice of the new unit ball has diameter 2, while the new unit ball has non-empty relatively weakly open subsets of arbitrarily small diameter. Consequently, the renormed space has LD2P and fails D2P.

Status

The answer to the arXiv:1206.3539 uncertainty is therefore:

$$\text{LD2P} \not\Longleftrightarrow \text{D2P}.$$

More precisely, D2P implies LD2P, but the converse fails, even within equivalent renormings of any Banach space containing an isomorphic copy of c_0 .

This is an already-known literature answer, not a new result from the run. The supporting paper explicitly records that it is answering the equivalence problem between slice/local D2P and D2P. The link to the exact arXiv:1206.3539 passage is by matching the definitions and the same question; the supporting paper cites the earlier Abrahamsen–Lima–Nygaard source for the problem.

Queue-signal note

The two deterministic signals that reserved arXiv:1206.3539 are same-paper false positives. The introduction asks for an analogue of a theorem and then proves it in the same paper. Later, the authors say they know no explicit reference for bidual-to-space inheritance of D2P, and then prove a stronger ai-ideal inheritance result in the same section. Neither passage should be counted as a promoted result.

References

- T. A. Abrahamsen, V. Lima, O. Nygaard, *Almost isometric ideals in Banach spaces*, arXiv:1206.3539.
- J. Becerra Guerrero, G. Lopez-Perez, A. Rueda Zoca, *Big slices versus big relatively weakly open subsets in Banach spaces*, arXiv:1304.4397; J. Math. Anal. Appl. 428 (2015), 855–865.